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Abstinence outcomes among women in reproductive health centers administered clinician or electronic brief interventions

Kimberly A. Yonkers[a,b,c,*], Jason I. Dailey[d], Kathryn Gilstad-Hayden[a], Steven J. Ondersma[e], Ariadna Forray[a], Todd A. Olmstead[f], Steve Martino[a,g]

[a] Department of Psychiatry, Yale School of Medicine, 300 George Street, New Haven, CT 06511, USA
[b] Department of Obstetrics, Gynecology and Reproductive Sciences, Yale School of Medicine, 333 Cedar Street, New Haven, CT 06510, USA
[c] Department of Epidemiology and Public Health, Yale School of Medicine, 60 College Street, New Haven, CT 06510, USA
[d] Uniformed Services University of the Health Sciences, 4301 Jones Bridge Road, Bethesda, MD 20814, USA
[e] Wayne State University, Merrill Palmer Skillman Institute, 71 E. Ferry Ave, Detroit, MI 48202, USA
[f] University of Texas at Austin, PO Box Y, Austin, TX 78713, USA
[g] VA Connecticut Healthcare System, 950 Campbell Avenue, West Haven, CT 06516, USA

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ABSTRACT

Treatment of substance use in women seeking reproductive healthcare is crucial for the health of both women and their offspring. Although abstinence from all substance use during pregnancy is optimal, it is difficult to achieve. This secondary analysis reports abstinence outcomes from a randomized clinical trial of screening, brief intervention, and referral to treatment (SBIRT) for substance use among women seeking reproductive healthcare services. Women who screened positive for substance use were randomly assigned to either clinician-administered SBIRT, an electronically-administered brief intervention (e-SBIRT), or an enhanced usual care condition. At a 6-month follow-up assessment, compared to enhanced usual care, the clinician-administered SBIRT increased 1-month point prevalence of abstinence from the primary substance by 7.7%, and e-SBIRT increased abstinence by 12.8%. Both brief interventions were more useful than enhanced usual care and could increase substance use abstinence rates among women in reproductive healthcare clinics. The electronic brief intervention is particularly attractive given that it appears as efficacious as, but requires fewer resources than, clinician-delivered brief interventions.

1. Introduction

Substance use among reproductive-age women is common, with past-year rates ranging from 7.5% for all illicit substance use to 21% for regular tobacco smoking and 23% for binge alcohol drinking. Administration (2019) Pregnancy influences substance use in that 60–80% of women will achieve abstinence from at least one substance in pregnancy (Forray, Merry, Lin, Ruger, & Yonkers, 2015; Hasin, Shmulewitz, & Keyes, 2019; Kendler, Ohlsson, Svikis, Sundquist, & Sundquist, 2017). While substance use treatment may further improve the likelihood of abstinence, treatment rates for substance use disorders (SUDs) were < 10% over the past decade (Creedon & Cook, 2016) and were lower for pregnant and non-pregnant women than men (Green, 2006; Martin, Longinaker, & Terplan, 2015).

Screening, Brief Intervention, and Referral to Treatment (SBIRT) is a collection of strategies involving: 1) universal screening, usually in a healthcare setting, to identify individuals who likely misuse substances and 2) a brief therapeutic intervention designed to reduce substance use and promote additional treatment seeking (Babor et al., 2007). Even though SBIRT is best established for the reduction of alcohol (Bertholet, Daeppen, Wietlisbach, Fleming, & Burnand, 2005; Whitlock et al., 2004) and tobacco use (Stead, Bergson, & Lancaster, 2008), other work (Aldridge, Dowd, & Bray, 2017; Bernstein et al., 2005) (*c.f. Lundahl* (Lundahl, Kunz, Brownell, Tollefson, & Burke, 2010) *and Halladay* (Halladay et al., *2019*)), including ours (Martino et al., 2018; Ondersma, Svikis, & Schuster, 2007), suggests that reductions in illicit substance use may be possible under some circumstances. Moreover, studies suggest that in some settings, women may benefit more than men from SBIRT (Blow et al., 2006; Saitz et al., 2009).

While evidence for SBIRT is positive, there are problems with its

* Corresponding author at: Department of Psychiatry, Yale School of Medicine, 300 George Street, New Haven, CT 06511, USA.
E-mail addresses: Kimberly.Yonkers@yale.edu (K.A. Yonkers), Jason.I.Dailey.mil@mail.mil (J.I. Dailey), Kathryn.Gilstad-Hayden@yale.edu (K. Gilstad-Hayden), s.ondersma@wayne.edu (S.J. Ondersma), Ariadna.Forray@yale.edu (A. Forray), TOlmstead@austin.utexas.edu (T.A. Olmstead), Steve.Martino@yale.edu (S. Martino).

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widespread dissemination (Hall, Staiger, Simpson, Best, & Lubman, 2016). The burden of initial training (Beich, Gannik, & Malterud, 2002), maintenance of proficiency (Hallgren, 2018; Schwalbe, Oh, & Zweben, 2014) and time required for its delivery (Aalto, Pekuri, & Seppa, 2003; Yarnall, Pollak, Ostbye, Krause, & Michener, 2003) are impediments to SBIRT implementation. Electronic delivery of an intervention is a modality that may address some of these issues (Harris & Knight, 2014; Ondersma et al., 2015; Ondersma, Chase, Svikis, & Schuster, 2005; Ondersma et al., 2007; Ondersma, Svikis, Thacker, Beatty, & Lockhart, 2013; Riper et al., 2014; Rooke, Thorsteinsson, Karpin, Copeland, & Allsop, 2010; Tansil et al., 2016). Electronic interventions also provide privacy, especially for behaviors that are stigmatized such as substance misuse (Verhoeks, Teunissen, van der Stelt-Steenbergen, & Lagro-Janssen, 2019). Prior work showed that electronically-delivered brief interventions were equivalent to clinician-delivered interventions in reduction of illicit substance use (Schwartz et al., 2014) and superior to enhanced usual care for reduction of substance use among women seeking reproductive health care (Martino et al., 2018).

The present study is a secondary analysis from a 3-group randomized controlled trial that compared an electronically-delivered SBIRT (e-SBIRT), a clinician-delivered SBIRT (SBIRT), and enhanced usual care (EUC)·(Martino et al., 2018). We previously reported greater reductions among SBIRT groups compared to EUC in days per month of use of the substance they designated as primary at baseline; herein we examine abstinence rates in the 6-month follow-up period after SBIRT administration and explore outcomes in pregnant and non-pregnant subgroups, as well as within primary substance subgroups. We hypothesized that both SBIRT and e-SBIRT would result in higher rates of abstinence from primary substance use than EUC in the overall cohort. We also sought to descriptively characterize intervention effects within subgroups defined by pregnancy status and primary substance designated at baseline.

2. Material and methods

2.1. Inclusion/exclusion

Participants were identified among pregnant and non-pregnant adult women who presented for routine outpatient reproductive healthcare in two urban academic hospital-based clinics. Inclusion criteria included: 1) a minimum substance-specific Alcohol, Smoking, and Substance Involvement Screening Test (ASSIST) (Humeniuk et al., 2008) score of 11 for alcohol in non-pregnant women or 6 for pregnant women, or 4 for either pregnant or non-pregnant women for any substance other than alcohol; 2) ability to speak English. Women were ineligible if they: 1) were currently or would imminently be incarcerated or hospitalized; 2) could not provide two collateral contacts; or 3) were currently engaged in treatment. The study was approved by an institutional review board and registered at clinicaltrials.gov (identifier: NCT01539525).

2.2. Treatment assignment

After meeting screening criteria and upon provision of written informed consent, participants completed a computerized assessment battery (see below). A research assistant (RA) then ran a computerized urn randomization program (the gRand urn randomization program, version 1.1) to determine treatment group assignment. Factors in the urn randomization included primary substance and pregnancy status. Participants who were assigned to the e-SBIRT condition continued with laptop administration of the intervention, after which they were given a pamphlet with treatment resources; SBIRT participants met with a clinician who administered the interview and provided the participant with a pamphlet. Those in the control condition (EUC) were told their level of risk and were given a pamphlet listing local treatments that could help them to cut down or stop their use. Follow-up was conducted at 1-, 3-, and 6-months post randomization.

2.3. Assessments

We collected study data via computer to the greatest extent possible to minimize any influence on treatment effects. At baseline this included demographic data, ASSIST scores, and presence of a substance use diagnosis according to the Mini-International Neuropsychiatric Inventory (version 5.0.0, Clinician-Rated) (Sheehan et al., 1998). The computer also asked participants to identify their most problematic substance among those used in the prior 3 months, and we labelled this as the "primary substance of abuse". At this visit and follow up visits, we electronically collected the Fagerstrom Test for Nicotine Dependence (Heatherton, Kozlowski, Frecker, & Fagerstrom, 1991), and RAs collected data on treatment and self-help contacts according to the Treatment Utilization Form (McLellan, Alterman, Cacciola, Metzger, & O'Brien, 1992).

Pregnancy status at baseline was determined via computer-based self-report and confirmed with medical record review for baseline and at the 1-, 3-, and 6-month follow-up visits.

An RA administered Timeline Follow-back (TLFB) interview (Fals-Stewart, O'Farrell, Freitas, McFarlin, & Rutigliano, 2000; Sobell & Sobell, 1992) to obtain information on all substances used during the 28 days prior to the baseline assessment and since the last assessment for follow-up interviews using calendar-based recall methods. A urine drug screen (UDS), collected at each visit, was assayed for amphetamines, barbiturates, benzodiazepines, buprenorphine, cocaine, methadone, methaqualone, opiates, oxycodone, phencyclidine, propoxyphene, cannabinoids, alcohol (EtG/EtS), and nicotine (cotinine) (Redwood Toxicology Labs; Santa Rosa, CA). Participants were compensated with gift cards for each study assessment completed, which totaled up to $250 per participant if all 4 assessments were completed. For the TLFB compared to UDS, the false negative rate (participant reported no use when there was a positive UDS) at one month was 0.04, 0.03 and 0.07 for EUC, e-SBIRT and SBIRT, respectively. At 3 months, rates were 0.06, 0.07 and 0.07 for EUC, e-SBIRT and SBIRT respectively. At 6 months the rates were 0.13, 0.11 and 0.11 for EUC, e-SBIRT and SBIRT, respectively. Accordingly, the TLFB was deemed an accurate reflection of substance use.

2.4. Conditions

The brief interventions in the SBIRT and e-SBIRT conditions were each approximately 20 min in length, were based on motivational interviewing principles (Miller & Rollnick, 2013), and were designed to mirror each other in overall structure. Both brief interventions sought simultaneously to enhance motivation to quit or reduce use, and to facilitate treatment involvement. Consistent with motivational interviewing principles, both the SBIRT and e-SBIRT interventions sought to express empathy and respect autonomy via open-ended questions, affirmations, reflections, and summaries; and sought to elicit change talk as well as specific change goals, in part through techniques such as personalized normed feedback.

The e-SBIRT included a talking animated narrator (a parrot) that delivered the intervention and provided feedback in a computer-adaptive format using the Computerized Intervention Authoring System (CIAS) (Ondersma et al., 2007). In addition to the narrator, the program asked questions and provided text answers and images such as pictures of items that could be earned with money saved from not buying substances. This is an unguided stand-alone intervention that does not require input from a therapist.

SBIRT therapists for this study included clinicians with backgrounds in nursing, social work, or medicine (obstetrics-gynecology). All were trained via a workshop that entailed 15 h of learning, followed by practice cases, coaching and monthly supervision. Clinician-

administered sessions were audio recorded and rated with the Independent Tape Rater Scale (Loree, Yonkers, Ondersma, Gilstad-Hayden, & Martino, 2019; Martino, Ball, Nich, Frankforter, & Carroll, 2008). Finally, the EUC control condition consisted of a 2-minute discussion of the participant's substance use and review of a handout listing local treatment resources. All interventions were delivered the same day as screening and randomization.

2.5. Statistical analyses

The primary outcome for this analysis was one-month point prevalence abstinence (i.e. no use in the previous 28 days) from the participant's primary substance, and it was measured at the one-, three- and six-month follow-up visits. Women were coded as abstinent if they reported no use of their primary substance over the past 28 days on the TLFB and screened negative for their primary substance on the UDS. For participants who were missing UDS results (n = 39 at Month One and n = 34 at Month Three), abstinence was based solely upon self-report in the TLFB. The outcome for women who were lost to follow-up or were missing both urine and self-report data (n = 17 at Month One, n = 32 at Month Three and n = 53 at Month Six) was imputed as not abstinent.

Mean and standard deviation for continuous variables, and frequency and percentage for categorical variables were calculated to describe the characteristics of women in the cohort. One-month point prevalence abstinence rates at the one-, three- and six-month follow-up visits were analyzed in separate exact (because of small cell sizes) logistic regression models to compare the odds of abstinence in SBIRT and e-SBIRT groups versus EUC, adjusting for pregnancy status (0 = not pregnant, 1 = pregnant) and primary substance (0 = nicotine, 1 = alcohol, 2 = illicit substance). Results were reported as odds ratios (OR) with 95% confidence intervals (CI). Holm's step-down multiple testing procedure was applied to control the type-one error rate given that we were comparing two treatment groups at three different time points.

Exploratory analyses (two-way contingency tables and Fisher's exact tests) were conducted to compare rates of abstinence between treatment groups among pregnancy and primary substance subgroups. To explore the potential impact of pregnancy, we recoded some women in the cohort as not pregnant because they delivered and were not pregnant at follow up visits, or as pregnant because they became pregnant during the study. We considered them as pregnant if that was their status during any part of the interval between visits. For exploratory analyses, $p < 0.05$ was considered statistically significant. All analyses were based on the intention-to-treat cohort and were performed with SAS version 9.4 (Cary, NC).

3. Results

3.1. Sample description

Details of participant flow are reported elsewhere (Martino et al., 2008) and are thus included in the Appendix. We screened 2421 women and randomized 439 who are included in the intention-to-treat analyses. A total of 383 women (87.2%) completed one-month follow-up, 371 women (84.5%) completed three-month follow-up, and 386 women (87.9%) completed six-month follow-up. The randomized sample had an overall mean age of 34.2 years, 66.7% were non-Hispanic African American, 66.3% were unemployed, and 18.2% reported being pregnant at baseline. Participants identified their primary substance used as: nicotine (57.2%), alcohol (11.6%), or illicit substances (31.2%). They used this primary substance an average of 23.7 days per month.

As shown in Table 1, women who were abstinent at the six-month follow-up were significantly younger than those who were not abstinent (30 vs 35 years old, $t = 2.4$, $p = 0.02$). Pregnant women were more likely than non-pregnant women to be abstinent at Month Three (18% versus 8%, respectively, $X^2 = 6.9$, $p = 0.01$) and Month Six (20% versus 8%, respectively, $X^2 = 7.7$, $p = 0.01$). Education was associated with abstinence at the one- and three-month follow-up visits, in that women with a high school degree but no college achieved the highest percentage of abstinence compared to either women without a high school degree or those with some schooling beyond high school (9% versus 2% and 3%, respectively, at Month One; and 15% versus 6% and 6%, respectively, at Month Three).

3.2. Primary outcomes

3.2.1. Overall abstinence

One-month point prevalence abstinence was identified in 5.5% (N = 24), 9.8% (N = 43), and 10.0% (N = 44) of women at Months One, Three and Six respectively. At each time point, raw rates of abstinence were higher among women in e-SBIRT (8.4%, 13.3%, and 16.1% at Months One, Three and Six, respectively) and SBIRT groups (7.6%, 11.0%, and 11.0% at Months One, Three and Six, respectively) than the EUC group (0.7%, 5.3%, 3.3% at Months One, Three and Six, respectively). Results from the exact logistic regression models are presented in Table 2. Independent of pregnancy status and primary substance, odds of one-month point prevalence abstinence were higher in the e-SBIRT and SBIRT groups compared to EUC at the one-month follow up visit ($OR_{e\text{-}SBIRT} = 12.9$, 95% CI = 1.8 to 562.1, $p = 0.003$ for e-SBIRT, and $OR_{SBIRT} = 11.4$, 95% CI = 1.6 to 498.6, $p = 0.007$ for SBIRT). Large odds ratios and wide confidence intervals reflect the low number of women achieving abstinence (n = 1) in the EUC group at Month One. Compared with EUC, the odds of abstinence at Month Three was over two times higher for e-SBIRT and SBIRT, but did not reach the significance threshold of $p < 0.025$ set by Holm's adjustment ($OR_{e\text{-}SBIRT} = 2.7$, 95% CI = 1.1 to 7.6, $p = 0.035$ for e-SBIRT and $OR_{SBIRT} = 2.1$, 95% CI = 0.8 to 5.9, $p = 0.148$ for SBIRT). At Month Six, women in the e-SBIRT group had over five times higher odds of abstinence than women in EUC (OR = 5.8, 95% CI = 2.0 to 20.2, $p < 0.001$); results were similar for the SBIRT group (OR = 3.7, 95% CI = 1.2 to 13.2, $p = 0.016$).

3.2.2. Abstinence stratified by pregnancy status

Pregnant women were more likely than non-pregnant women to be abstinent at each time point, although this was non-significant after adjustment for multiple comparisons independent of treatment group and primary substance (Month One OR = 3.0, 95% CI = 1.0 to 8.5, $p = 0.047$); Month Three (OR = 3.2, 95% CI = 1.4 to 7.1, $p = 0.005$); and Month Six (OR = 3.0, 95% CI = 1.3 to 6.8, $p = 0.01$).

3.2.3. Abstinence stratified by substance

At Month One and independent of treatment group and pregnancy status, abstinence rates were higher among women who drank or used an illicit substance compared to tobacco users ($OR_{alcohol} = 5.2$, 95% CI = 1.3 to 20.6, $p = 0.019$; $OR_{illicit} = 3.4$, 95% CI = 1.6 to 7.7, $p = 0.025$). Illicit substance users also had a significantly higher abstinent rate compared to tobacco users at Month Three ($OR_{illicit} = 1.5$, 95% CI = 0.7 to 3.3, $p = 0.001$). There was no difference between users identifying primary substances of alcohol and tobacco at three months. At the six-month follow-up, odds of abstinence did not differ by any of the primary substance groups.

3.3. Descriptive outcomes

Results from the descriptive subgroup analyses are presented in Fig. 1 for pregnancy subgroups, and in Fig. 2 for primary substance subgroups. Abstinence rates were highest among pregnant women in e-SBIRT at one, three and six-months post-intervention (16%, 25%, and 37%, respectively) compared to SBIRT (13%, 14%, and 11%) and EUC (0%, 15%, 14%), although no significant treatment effects were found among pregnant women at any time point. Among the primary

Table 1
Characteristics of randomized participants by 1-month point prevalence abstinence from primary substance at 1, 3, and 6-month follow-up visits.

Characteristic	Month 1		Month3		Month 6	
	Not abstinent (N = 415)	Abstinent (N = 24)	Not abstinent (N = 396)	Abstinent (N = 43)	Not abstinent (N = 395)	Abstinent (N = 44)
Age in years, mean (SD)[x]	34 (11)	33 (11)	34 (11)	33 (11)	35 (11)	30 (9)
Race/ethnicity, N (%)						
African-American, non-Hispanic	275 (66)	18 (75)	261 (66)	32 (74)	263 (67)	30 (68)
Hispanic-American	62 (15)	3 (13)	60 (15)	5 (12)	58 (15)	7 (16)
Caucasian-American, non-Hispanic	56 (13)	2 (8)	55 (14)	3 (7)	53 (13)	5 (11)
Other race/ethnicity[b]	22 (5)	1 (4)	20 (5)	3 (7)	21 (5)	2 (5)
Marital status, N (%)						
Married/living with partner	188 (45)	9 (38)	180 (45)	17 (40)	176 (45)	21 (48)
Other living situation	227 (55)	15 (63)	216 (55)	26 (60)	219 (55)	23 (52)
Pregnant, N (%)[c,+ x]	78 (19)	8 (33)	65 (16)	14 (33)	48 (12)	12 (27)
Education, N (%)[d,*,+]						
Some high school or less	143 (35)	3 (13)	137 (35)	9 (21)	135 (34)	11 (26)
High school graduate	164 (40)	17 (74)	154 (39)	27 (63)	157 (40)	24 (56)
Beyond high school	107 (26)	3 (13)	103 (26)	7 (16)	102 (26)	8 (19)
Employment status, N (%)						
Full time	62 (15)	4 (17)	62 (15)	4 (17)	60 (15)	6 (14)
Part time	74 (18)	8 (33)	74 (18)	8 (33)	71 (18)	11 (25)
Not working	279 (67)	12 (50)	279 (67)	12 (50)	264 (67)	27 (61)
Days/month using primary substance, mean (SD)[*,+]	24 (8)	18 (9)	24 (7)	18 (9)	24 (8)	22 (9)
Primary substance[*,+]						
Alcohol	45 (11)	6 (25)	44 (11)	7 (16)	45 (11)	6 (14)
Illicit substance[a]	126 (30)	11 (46)	116 (29)	21 (49)	121 (31)	16 (36)
Tobacco	244 (59)	7 (29)	236 (60)	15 (35)	229 (58)	22 (50)
Treatment group, N (%)[*, x]						
Enhanced usual care	150 (36)	1 (4)	143 (36)	8 (19)	146 (37)	5 (11)
e-SBIRT	131 (32)	12 (50)	124 (31)	19 (44)	120 (30)	23 (52)
SBIRT	134 (32)	11 (46)	129 (33)	16 (37)	129 (33)	11 (36)

Abbreviations: SD, standard deviation; e-SBIRT, electronically-delivered Screening, Brief Intervention, and Referral to Treatment; SBIRT, clinician-delivered Screening, Brief Intervention, and Referral to Treatment; **EUC**, Enhanced Usual Care; **ASSIST**, Alcohol, Smoking, and Substance Involvement Screening Test.
Note: column percentages may not add to 100 due to rounding.
[*] Significant at the 0.05 level for abstinence at month 1.
[+] Significant at the 0.05 level for abstinence at month 3.
[x] Significant at the 0.05 level for abstinence at month 6.
[a] Includes cannabis (n = 90), cocaine (n = 38), opiates (n = 4), phencyclidine (n = 3), benzodiazepines/sedatives (n = 2).
[b] Includes multiracial (n = 16), Native American (n = 5), Pacific Islander (n = 1), and Romanian (n = 1).
[c] Pregnant at 1, 3, or 6-month follow-up visit, respectively, not baseline.
[d] N = 2 women missing education data.

Table 2
Exact odds ratios predicting 1-month point prevalence abstinence from primary substance at 1, 3, and 6-month follow-up visits for e-SBIRT and SBIRT vs usual care, adjusted for pregnancy status and primary substance (N = 439).

Parameter	1-month follow up			3-month follow up			6-month follow up		
	Exact OR	(95% CI)	p-Value	Exact OR	(95% CI)	p-Value	Exact OR	(95% CI)	p-Value
e-SBIRT vs. EUC	12.9	(1.8, 562.1)	0.003	2.7	(1.1, 7.6)	0.035	5.8	(2.0, 20.2)	< 0.001
SBIRT vs. EUC	11.4	(1.6, 498.6)	0.007	2.1	(0.8, 5.9)	0.148	3.7	(1.2, 13.2)	0.016
e-SBIRT vs. SBIRT	1.1	(0.4, 3.0)	0.953	1.3	(0.6, 2.9)	0.609	1.6	(0.7, 3.4)	0.267
Pregnant vs. not pregnant	3.0	(1.0, 8.5)	0.047	3.2	(1.4, 7.1)	0.005	3.0	(1.3, 6.8)	0.010
Alcohol vs. tobacco	5.2	(1.3, 20.6)	0.019	2.9	(0.9, 8.5)	0.077	1.3	(0.4, 3.8)	0.741
Illicit substance vs. Tobacco	3.4	(1.1, 11.0)	0.025	3.4	(1.6, 7.7)	0.001	1.5	(0.7, 3.3)	0.305
Alcohol vs. illicit substance	1.5	(0.4, 5.1)	0.624	0.8	(0.3, 2.3)	0.915	0.9	(0.3, 2.6)	0.992

Abbreviations: OR, odds ratio; CI, confidence interval; e-SBIRT, electronically-delivered Screening, Brief Intervention, and Referral to Treatment; SBIRT, clinician-delivered Screening, Brief Intervention, and Referral to Treatment; **EUC**, Enhanced Usual Care.

substance subgroups, a treatment effect was seen only in the illicit substance group at Months-Three and Six, with the e-SBIRT seeing the highest rate of abstinence (25.0% at Months Three and Six), followed by SBIRT (16.3% and 12.4%, at Months Three and Six, respectively) and then EUC (6.3% at Month Three and 0.0% at Month Six). No treatment effect was found among the alcohol primary substance group, but the percentage of women achieving abstinence from alcohol was consistently highest among SBIRT (27% at Month One, 27% at Month Three, and 20% at Month Six) compared to e-SBIRT (9%, 9% and 13%, at Months One, Three, and Six, respectively) and EUC (0%, 8% and 0% at Months One, Three, and Six, respectively). A trend (p = 0.74) was seen among the nicotine primary substance subgroup, with 4% of women in e-SBIRT and 4% of women in SBIRT women achieving abstinence versus 0% in EUC. In the later follow-up visits, rates of

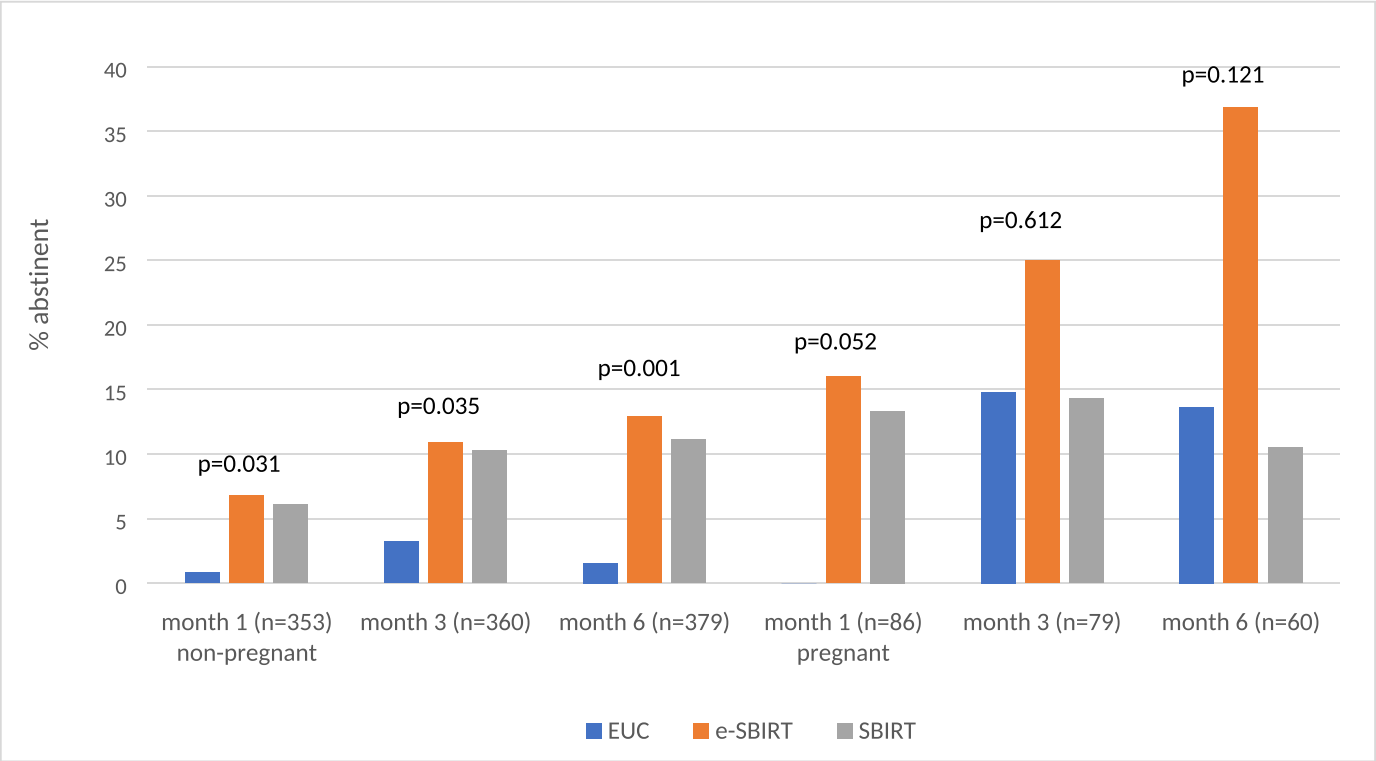

Fig. 1. One-month point prevalence abstinence at 1, 3, and 6-month follow-up visits by treatment group and pregnancy status (n = 439).

abstinence from nicotine was highest among the e-SBIRT group, but the treatment effect was not significant.

Rates of continuous abstinence throughout the study were low in all groups but were higher in the e-SBIRT and SBIRT groups compared to EUC. No women in the EUC group were continually abstinent, whereas continuous abstinence at six months was observed in a small percentage of women in the e-SBIRT (4%) and SBIRT (3%) groups (Fisher's Exact Test p = 0.023).

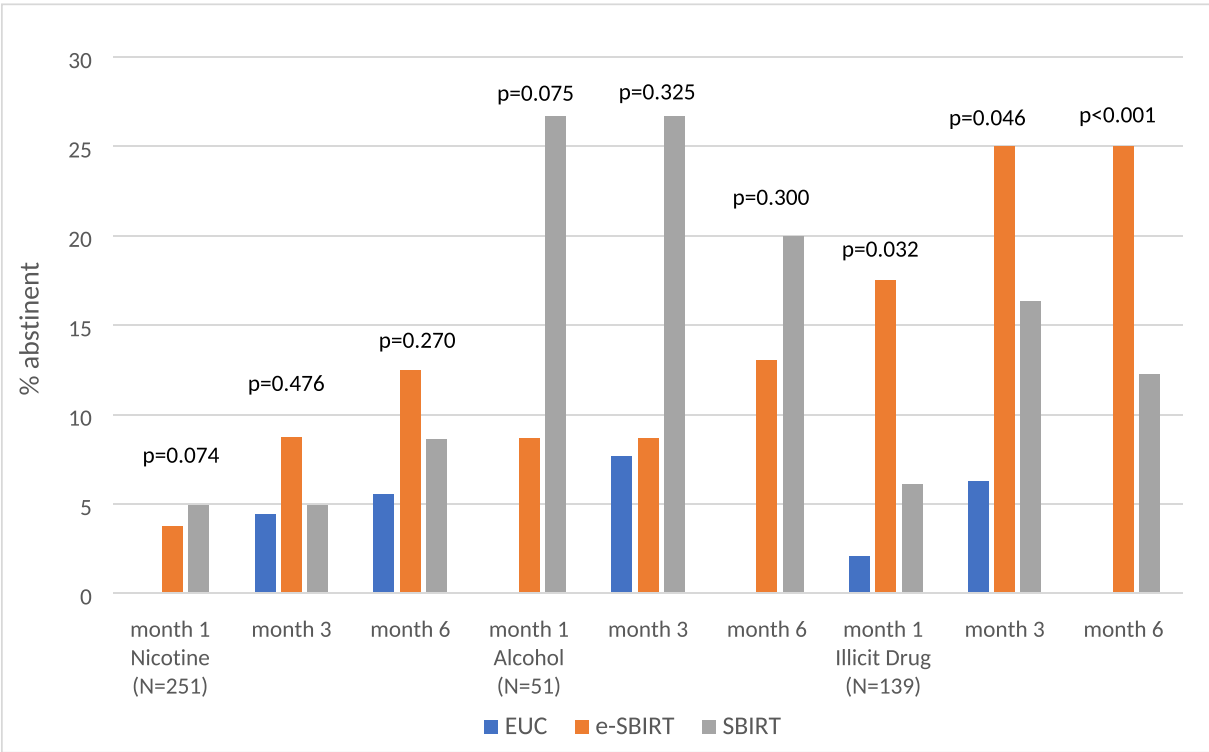

Fig. 2. One-month point prevalence abstinence at 1, 3, and 6-month follow-up visits by treatment group and primary substance (N = 439) Appendix.

4. Discussion

Abstinence is a particularly important outcome to assess in reproductive-aged women at risk of becoming pregnant, and pregnant women, since exposure to hazardous substances in pregnancy can have adverse birth consequences. This secondary analysis supports the superiority of both the e-SBIRT and SBIRT interventions over EUC in the achievement of abstinence from a woman's primary substance of misuse. This superiority was maintained at six months wherein the likelihood of abstinence was over five-fold higher in the e-SBIRT group and three-fold higher in the SBIRT group, compared to EUC. Although the trial was not powered to detect significant differences between the two brief interventions, e-SBIRT showed a notable efficacy advantage among pregnant women at the six-month outcome that should be examined in further trials. Our past work also demonstrates that e-SBIRT shows advantages in cost-effectiveness from both clinic and patient perspectives (Olmstead et al., 2019).

One month after the intervention, women who identified nicotine as their primary substance were less likely to attain abstinence compared to women who used alcohol and illicit substances, independent of treatment and pregnancy status. However, these differences were not seen at the end of the six-month follow-up period. There is little work comparing the likelihood of abstinence between various substances. In an earlier study, we also showed lower rates of abstinence from tobacco compared to other substances (cocaine, marijuana and alcohol) among pregnant women (Forray et al., 2015). Similarly, another longitudinal study in pregnancy showed lower rates of abstinence from nicotine or marijuana cigarettes compared to cocaine, 3,4-methylenedioxy-*N*-methylamphetamine, or alcohol (Moore et al., 2010). It may be that nicotine is especially addictive, even compared to other hazardous substances. Alternatively, or in conjunction with this possibility, the ease of access to cigarettes or the fact that they are licit may contribute to continued use. Alcohol is also licit but there has been a major public health campaign to minimize or cease alcohol use in pregnancy.

In our study, and for the primary substance subgroup analysis, we found a treatment effect at Months Three and Six for illicit substances wherein e-SBIRT performed better than both SBIRT and EUC. This finding is consistent with another trial using e-SBIRT interventions like the one used in this study, which found significantly decreased illicit substance use among postpartum women (Ondersma, Svikis, Thacker, Beatty, & Lockhart, 2014). The interactive nature of the e-SBIRT program and its use of tailored content, inclusion of imagery, and personalized feedback about issues salient to women may have contributed to e-SBIRT's efficacy with illicit substance use. Likewise, the structured e-SBIRT approach may have offered more certainty that key intervention components (e.g., summarizing and supporting what the participant chose to do) were delivered, thereby better safeguarding intervention adherence and rendering an effect (Loree et al., 2019). An additional factor is that the computer intervention may be preferable to women who about possible disapproving attitudes from live counselors (Verhoeks et al., 2019). This may predispose participants to feel comfortable with and engage with the computer intervention, and they may do so differentially for illicit substances vs. alcohol. In contrast, SBIRT showed potential for an advantage (vs. e-SBIRT) in reducing alcohol use.

Previous work shows the benefit of an e-SBIRT for substance misuse (Harris & Knight, 2014; Ondersma et al., 2015; Ondersma et al., 2005; Ondersma et al., 2007; Ondersma et al., 2013; Riper et al., 2014; Rooke et al., 2010; Tansil et al., 2016). An e-SBIRT also has structural advantages, including lower costs, less time spent by providers in training and treatment, and more reliable delivery of treatment (Pollick et al., 2015). This renders the e-SBIRT a compelling treatment tool in reproductive healthcare settings. Such options may be particularly helpful in non-psychiatric settings where computer administration can obviate the need to hire a clinician who can administer an SBIRT.

As expected, pregnant women were more likely than non-pregnant women to attain abstinence (Hasin et al., 2019; Kendler et al., 2017). Even though abstinence is a particularly salient outcome for pregnant women, our results tended to show the strongest overall treatment differences for non-pregnant women in the study (see Fig. 1). This finding may be due to low abstinence rates with EUC in the non-pregnant cohort throughout the study. Pregnancy is a powerful motivator for abstinence (Forray et al., 2015). Also, the pregnant subgroup was smaller and although there were substantial differences among groups, a larger sample size was likely needed to overcome the powerful spontaneous abstinence rates typically seen in pregnancy. These findings speak to the need to better understand the processes that underlie the achievement of spontaneous abstinence in pregnancy.

This study has numerous strengths, including the inclusion of pregnant and non-pregnant women, accessing women in a reproductive health setting, and good retention through the 6-month follow-up. However, it should be noted that this study was conducted within urban academic healthcare settings with a predominantly minority population, thereby limiting the findings' generalizability. In addition, study personnel provided most SBIRT sessions, rather than clinic staff. Hence, the magnitude of the effect of SBIRT when used by reproductive healthcare providers and staff remains to be determined. Further, TLFB interviews for this study were conducted by RAs who may not have been completely blind to participant intervention status. However, urine testing showed high agreement with self-report, suggesting that participant under-reporting and RA bias, if present, were minimal. In addition, cell sizes for many analyses were small, meaning that many comparisons were under-powered and the influence of random variation was increased. Similarly, the study was not powered to compare the two brief interventions to each other or to examine intervention effects based on pregnancy status.

5. Conclusions

Although abstinence from the primary substance was a rare outcome in this sample, clinician-delivered SBIRT and e-SBIRT both produced modest and statistically significant effects in promoting abstinence from substances. Nonetheless, the relatively low cost and ease of implementation for the e-SBIRT intervention (Olmstead et al., 2019) suggests that it could be a practical and acceptable method for implementing brief interventions at scale for at-risk women. In addition, large-scale implementations could support ongoing optimization efforts, which would be facilitated by the ease with which electronic interventions can be modified, and with which improved versions can be disseminated. Further effort in this area appears warranted.

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Disclaimer

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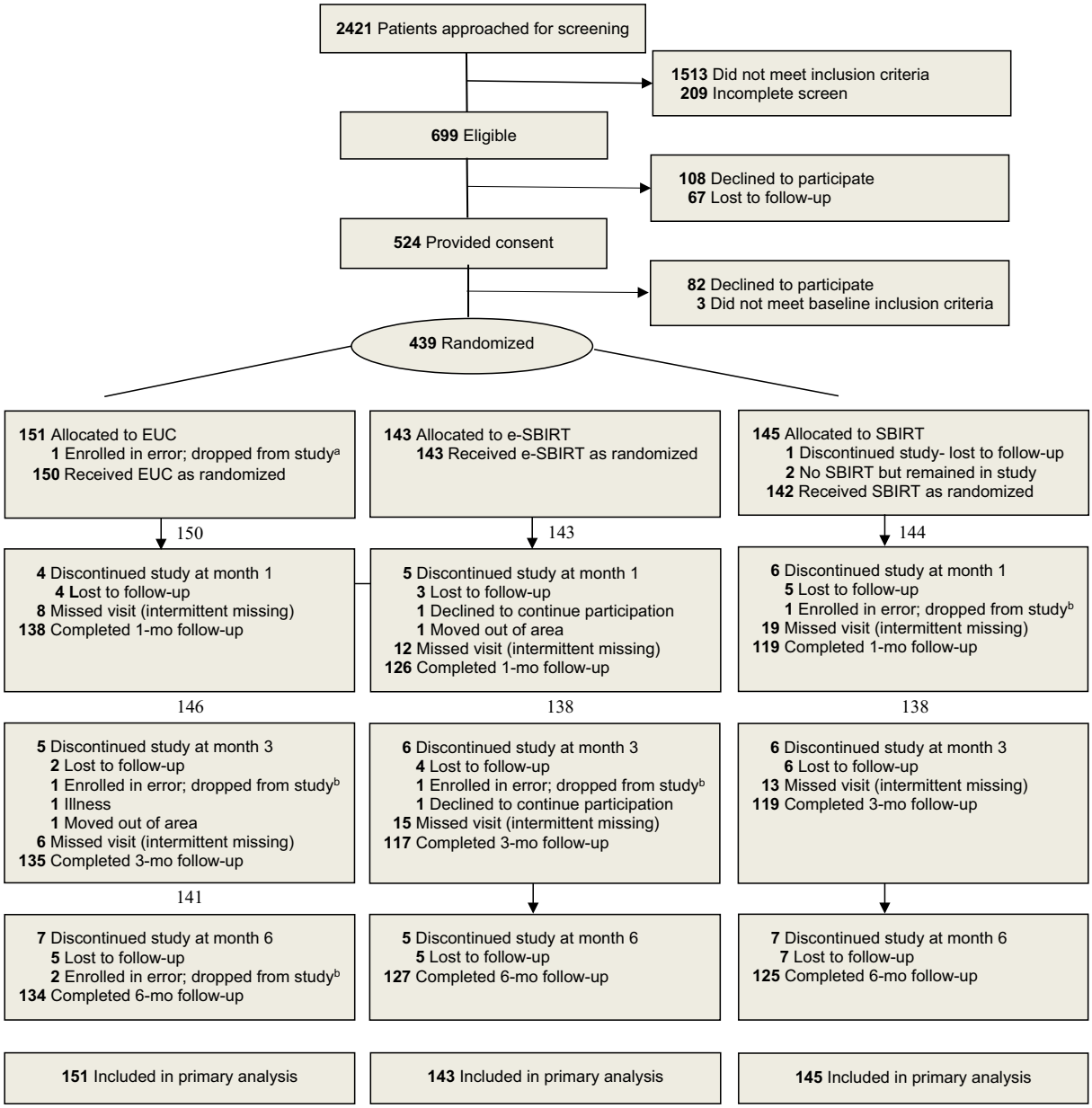
CRediT authorship contribution statement

Kimberly A. Yonkers: Conceptualization, Methodology, Writing - review & editing. **Jason I. Dailey:** Writing - review & editing. **Kathryn Gilstad-Hayden:** Data curation, Formal analysis. **Steven J. Ondersma:** Conceptualization, Methodology, Software, Writing - review & editing. **Ariadna Forray:** Writing - review & editing. **Todd A. Olmstead:** Writing - review & editing. **Steve Martino:** Conceptualization, Methodology, Writing - review & editing.

Declaration of competing interest

Dr. Yonkers discloses royalties from Up To Date; Dr. Ondersma discloses that he is part-owner of Interva, Inc., which markets the intervention authoring tool that was used to develop the electronic intervention for this study.

Appendix A



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